In [5]:

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import norm
from collections import Counter

from sklearn.datasets import load_iris
iris_data = load_iris()
df = pd.DataFrame(data=iris_data.data, columns=iris_data.feature_names)
df['species'] = pd.Categorical.from_codes(iris_data.target, iris_data.target_names)

print("Dataset Overview:")
df.info()
print("\nSummary Statistics:")
df.describe()
```

```
Dataset Overview:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   sepal length (cm)  150 non-null    float64
 1   sepal width (cm)   150 non-null    float64
 2   petal length (cm)  150 non-null    float64
 3   petal width (cm)   150 non-null    float64
 4   species            150 non-null    category
dtypes: category(1), float64(4)
memory usage: 5.1 KB

Summary Statistics:
```

Out[5]:

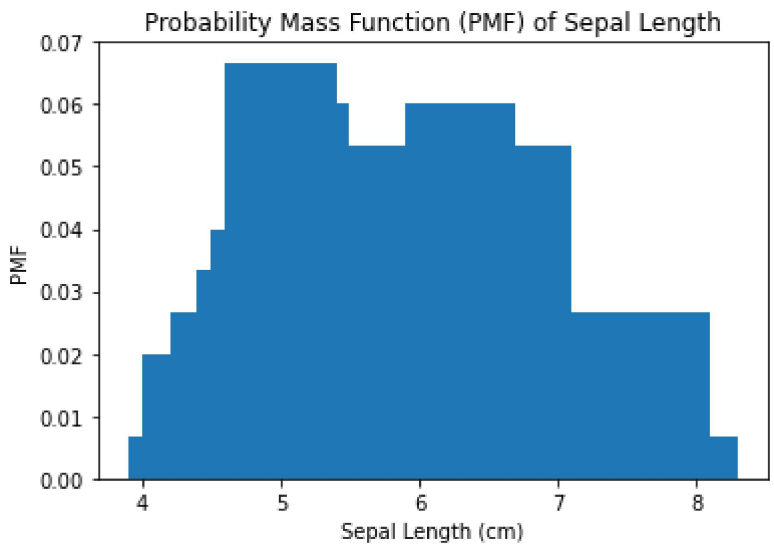
	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
count	150.000000	150.000000	150.000000	150.000000

PMF

In [8]:

```
sepal_length = df['sepal length (cm)']
def calculate_pmf(data):
    counts = Counter(data)
    total_count = len(data)
    pmf = {k: v / total_count for k, v in counts.items()}
    return pmf

pmf = calculate_pmf(sepal_length)
plt.bar(pmf.keys(), pmf.values())
plt.xlabel('Sepal Length (cm)')
plt.ylabel('PMF')
plt.title('Probability Mass Function (PMF) of Sepal Length')
plt.show()
```

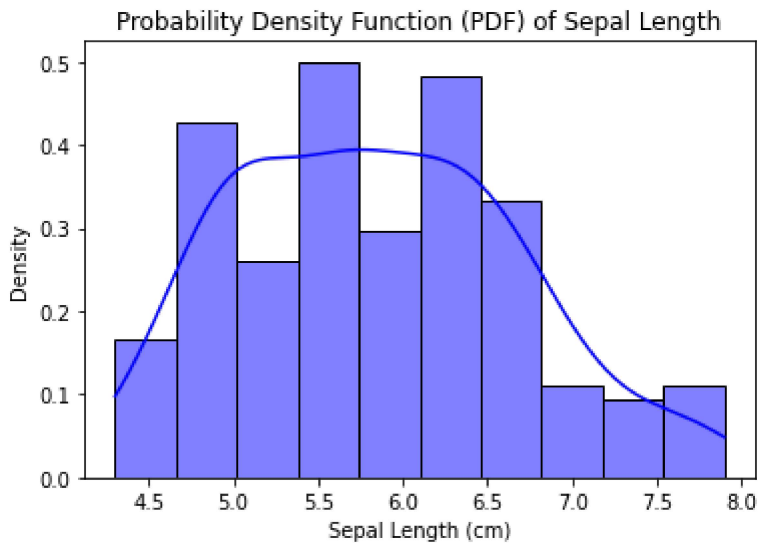


PDF

In [9]:

```
sns.histplot(sepal_length, kde=True, stat="density", bins=10, color="blue")
plt.title('Probability Density Function (PDF) of Sepal Length')
plt.xlabel('Sepal Length (cm)')
plt.ylabel('Density')
plt.show()
```

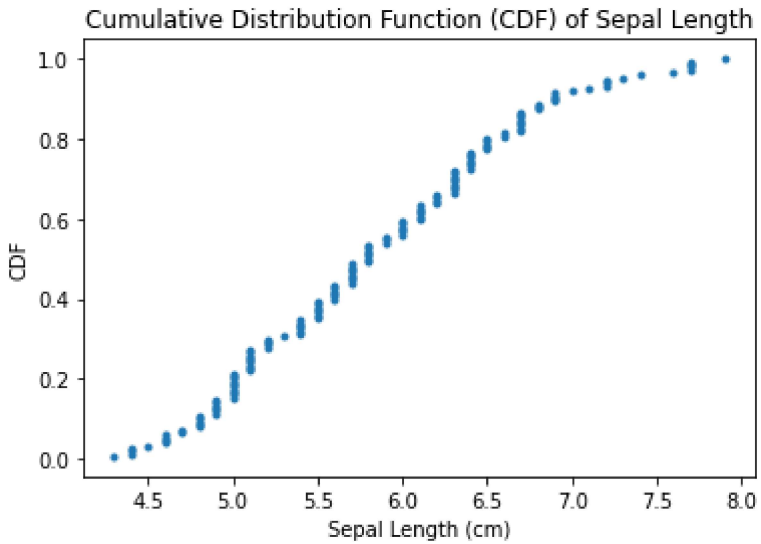
Out[9]: Text(0, 0.5, 'Density')



CDF

In [10]:

```
sorted_data = np.sort(sepal_length)
cdf = np.arange(1, len(sorted_data) + 1) / len(sorted_data)
plt.plot(sorted_data, cdf, marker='.', linestyle='none')
plt.xlabel('Sepal Length (cm)')
plt.ylabel('CDF')
plt.title('Cumulative Distribution Function (CDF) of Sepal Length')
plt.show()
```



Summary Statistics

In [11]:

```
mean_value = np.mean(sepal_length)
std_dev_value = np.std(sepal_length)
min_value = np.min(sepal_length)
max_value = np.max(sepal_length)
print(f"Mean: {mean_value}")
print(f"Standard Deviation: {std_dev_value}")
print(f"Minimum: {min_value}")
print(f"Maximum: {max_value}")
```

```
Mean: 5.843333333333335
Standard Deviation: 0.8253012917851409
Minimum: 4.3
Maximum: 7.9
```